

Sriharsha Annamaneni

sriharsha0806@gmail.com | (+91) 798 178 7689 | sriharsha0806.github.io/about

Applied AI/ML engineer with 7+ years shipping production AI across healthcare, insurance, automotive, and enterprise platforms. Current Senior Applied AI/ML at JPMC, focused on evaluation alignment and agents for long-horizon tasks. Recent work includes enterprise RAG, LangGraph and MCP agent workflows, Text2SQL systems, model fine-tuning, and production LLM serving infrastructure.

Work Experience

Senior Applied AI/ML | Jun 17, 2026 -- Present

JPMC, Bengaluru

- Working on evaluation alignment and long-horizon agent tasks for applied AI/ML systems.

Senior Gen AI Data Scientist | May 2024 -- Jun 15, 2026

Fractal, Bengaluru

- Health Insurance SOP Assistant.** Built and scaled an enterprise RAG platform used by 5K+ daily users, with hybrid retrieval, Azure Functions + Service Bus ingestion, Azure AD access control, 90% accuracy, sub-second latency, and \$11M+ in credited operational savings.
- Improved response quality 40% over the initial UAT baseline with DSPy evaluator-optimizer loops, SME feedback, GEPA prompt/program optimization, and Best-of-N sampling for sensitive query types.
- Built a distributed token caching and rate-limiting service in FastAPI + Redis with Lua scripts to coordinate shared OpenAI TPM quotas across parallel workers.
- Text2SQL Chatbot.** Led a LangChain + LangGraph multi-agent Text2SQL system with Azure AI Search retrieval, PII/hallucination guardrails, AKS deployment, and a 27% SQL generation accuracy improvement.
- Appeals, enrollment, and email automation.** Built LangGraph and MCP-powered reasoning workflows with SQL Server, MongoDB, Kafka, validation agents, fallout classification, supplier communication drafts, and human escalation paths.

Senior AI Engineer | May 2021 -- Apr 2024

Bosch, Bengaluru

Built ML tooling for defect triage, test planning, and in-cabin safety systems across automotive product lines.

- Ditto.** Built semantic similarity and RAG search over historical defects with sentence-transformers, Qdrant, dense + BM25 scoring, and cross-encoder reranking; time-to-resolution dropped roughly 30%.
- Interior Monitoring.** Built seat-belt and drowsiness detection models with GradCam++ failure analysis and evidential deep learning abstention; error rate dropped 40%.
- Bug-to-Test Linkage.** Fine-tuned MPNet on bug-test pairs and integrated traceability into the test management system; throughput improved 75x and effective coverage reached 95%.

Computer Vision Engineer | Oct 2020 -- Apr 2021

Aimlytics, Hyderabad

- Built an automated speech dubbing pipeline chaining ASR, speaker diarization, and TTS for multilingual media workflows.
- Adapted a two-stage Indic TTS stack using a Tacotron-style acoustic model and WaveNet vocoder on IIT Madras Indic TTS data.

Research Engineer | Nov 2019 -- Jun 2020

Sirena Technologies, Bengaluru

- Developed an offline wake-word model using GRUs for embedded consumer devices, cutting detection latency by 13%.
- Trained a Siamese network for face recognition that reached 99.8% accuracy on the internal evaluation set.

Core Skills

- GenAI and Agents:** LangChain, LangGraph, DSPy, Semantic Kernel, CrewAI, PydanticAI, MCP, A2A, Ollama
- ML / DL:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, OpenCV, Optuna
- Training and Serving:** QLoRA, distributed data loading, optimizer-state sharding, activation checkpointing, vLLM, SGLang, TensorRT-LLM
- Backend and Data:** FastAPI, Streamlit, Redis, MongoDB, Azure SQL, Cosmos DB, PySpark, Databricks, Snowflake, Azure Data Factory
- Cloud and MLOps:** Azure, AWS, Vertex AI, Kubernetes (AKS), Docker, MLflow, DVC, LangSmith, Arize AI, Grafana
- Languages:** Python, C++, Go

Research Experience

Research Fellow | Apr 2017 -- Jun 2019

IIIT Hyderabad, Hyderabad

- Applied deep learning research under Prof. C.V. Jawahar on road-audit systems and autonomous navigation, focused on efficient semantic segmentation for field deployment.
- Implemented and optimized PSPNet, ERFNet, MobileNet, and DeepLabV3 in PyTorch; several went into pilot deployments.
- Co-authored *Efficient Semantic Segmentation using Gradual Grouping*, presented at the CVPR Workshop 2018, where it received Best Runner-up.

Education

- **M.S. in Electrical Engineering**, Florida Institute of Technology | 2016
- **B.E. in Electronics and Communication Engineering**, Manipal Institute of Technology | 2014

Certifications and Reviewing

- *Scratch to Scale: Large-Scale Training in the Modern World* (Maven, 2025). Fine-tuned Mistral 7B on MedInstruct-52k with QLoRA across 4xA100s using DDP and activation checkpointing; LLM-as-judge win rate 78% vs 61% against base Mistral-7B on 500 held-out medical instructions.
- Technical reviewer for GenAI/ML books including *AI Agents in Action*, *AI Apps with LangChain*, *Generative AI Applications with AWS*, *Optimization Bootcamp*, and *Context Engineering with DSPy*.